

Puvanakopis Mehanathan

Intern AI/ML Engineer

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About Me

AI/ML-focused Software Engineering undergraduate experienced in LLMs, RAG systems, and intelligent applications. Skilled in designing and building scalable AI-powered and full-stack web applications using modern frameworks and APIs. Passionate about solving real-world problems through data-driven and intelligent systems. Eager to contribute to impactful projects while continuously improving technical and research skills.

Education

BSc(Hons) Software Engineering

Sabaragamuwa University of Sri Lanka

2023 – Present

GPA: 3.37 / 4.00

- Specialization: RAG, AI Agents & Web Development
- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Web Development, Software Engineering, Generative AI, Data Structures & Algorithms

Experience

AI Developer

ZENCORD (Freelance)

Aug 2025 – April 2026

- Built and deployed AI solutions using LLMs, RAG pipelines, and agent-based systems for real-world applications.
- Developed scalable backends with FastAPI and vector databases (ChromaDB, Pinecone) for semantic search and chatbots.
- Created end-to-end data pipelines and delivered AI applications including e-commerce assistants and data analysis tools.

Full Stack Engineer

Fiverr (Freelance)

Feb 2024 – Dec 2024

- Developed responsive full-stack web applications using React.js and Next.js with backend development in Node.js and Express.js.
- Built REST APIs, authentication systems, and scalable application features for modern web platforms.
- Optimized UI performance and delivered client-focused solutions across multiple domains using modern development practices.

Skills

- **Programming Languages:** Python, JavaScript, Java, C
- **Databases:** MySQL, MongoDB, PostgreSQL, ChromaDB, Pinecone
- **AI / ML:** LLMs, RAG Pipelines, AI Agents, Prompt Engineering, Embeddings, Semantic Search, Hugging Face Transformers
- **Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Data Cleaning, Data Visualization, EDA
- **Web & API Development:** React, Next.js, Tailwind CSS, FastAPI, Node.js, Express.js, Flask, REST APIs
- **Other Tools:** Git, GitHub, Docker, Web Scraping, API Integration

Projects

E-Commerce AI Agent | Conversational AI Assistant for Online Store Apr 2026 – May 2026

- *Overview:* AI-powered e-commerce assistant that interacts with a live SQLite database for product browsing, orders, and support.
- *AI Features:* LangGraph-based AI agent with tool calling, memory, and context-aware responses.
- *Tech Stack:* Python, LangGraph, LangChain, ChatGroq, SQLite, Pydantic

CampusEase | University Accommodation & Vehicle Rental Platform Apr 2025 – Jul 2025

- *Overview:* Full-stack platform for booking rooms and vehicles with role-based access for students, staff, and owners.
- *AI Features:* RAG-based chatbot using ChromaDB for smart, context-aware responses.
- *Tech Stack:* React, FastAPI, MongoDB, ChromaDB, Groq API, RAG

Smart Attendance System | AI Face Recognition & QR-Based Attendance Mar 2026 – May 2026

- *Overview:* University-focused mobile attendance system using QR codes with real-time student identity verification and location tracking.
- *AI Features:* Face recognition using OpenCV with geolocation validation to reduce proxy attendance.
- *Platform:* Built a cross-platform mobile app for students and lecturers using React Native.
- *Tech Stack:* React Native, Python, OpenCV, Node.js, MongoDB, REST APIs

Social Media Health Analysis | Data Analysis & ML Project Mar 2026 – Apr 2026

- *Overview:* Analyzed impact of social media usage on mental and physical health indicators.
- *AI Features:* EDA, statistical analysis, and visualization to identify behavioral patterns.
- *Tech Stack:* Python, Pandas, NumPy, Matplotlib, Seaborn

Certificates

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| • IBM RAG and Agentic AI Professional Certificate
Coursera IBM | Link |
| • IBM AI Developer Professional Certificate
Coursera IBM | Link |
| • Google Advanced Data Analytics Professional Certificate
Coursera Google | Link |
| • Crash Course on Python
Coursera Google | Link |

Volunteering

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| IEEE Student Branch of SUSL
Active member | <i>Member</i> |
| IEEE WIE Student Branch Affinity Group of SUSL (PearlHack)
March 2025 | <i>Finance Team</i> |
| IEEE Student Branch of SUSL
February, 2025 | <i>Program Team Member</i> |

References

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